

# Engineering Design Report

Fractional Forge — Project a\_compact\_commercial\_airsource\_heat\_pump

Generated: 2026-05-05

## Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£30.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

8

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

### Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction

[E] LLM assumption or unverified hypothesis

# Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

| Check                              | Status | Reason                        | Evidence     |
|------------------------------------|--------|-------------------------------|--------------|
| 1. Market TAM/SAM/SOM Defined      | FAIL   | Missing market data           | Not computed |
| 2. Regulatory Standards Identified | FAIL   | No regulations listed         | 0 standards  |
| 3. Cost Ceiling Specified          | PASS   | Target provided               | £30.00       |
| 4. Spatial Envelope Provided       | PASS   | Volume established            | 75 m³        |
| 5. Modules Decomposed              | PASS   | System architecture built     | 8 modules    |
| 6. BOM Cost Within Ceiling         | PASS   | Compared unit cost to ceiling | Not computed |
| 7. Risk Matrix Saturated           | PASS   | FMEA rows generated           | Risks exist  |

### Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

### Data Sources

[System] Internal gate checks

Overall Section Grade: A

# 1. Brief & Requirements [A]

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## 1.1 Mission & Use Case

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Mission

Not computed

Use Case

Not computed

## 1.2 Market Context & Timing [B]

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Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

# 1.3 Competitor Landscape [C]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Not computed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                               |
| Product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Ecodan CAHV-R450YA (and related Hydrodan commercial range)                                                                                                                                                                                    |
| Pricing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Industry sources suggest installed costs for comparable Mitsubishi commercial hydronic heat pumps in the 30–40kW range typically exceed £8,000–£12,000 per unit at point of sale to installer, though list pricing is not publicly published. |
| Tech Specs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Not computed                                                                                                                                                                                                                                  |
| <div><div>STRENGTHS</div><div>Established MCS certification and extensive UK installer network,Strong brand recognition and after-sales support infrastructure,Modular architecture allows flexible capacity matching,Comprehensive controls integration with BMS protocols</div></div> <div><div>WEAKNESSES</div><div>R32 refrigerant faces increasing regulatory pressure under F-Gas phase-down,Higher price point potentially excludes smaller commercial buyers,Product range optimised for larger installations; sub-30kW niche less well served,Split hydronic architecture less prominently marketed than VRF alternatives in their portfolio</div></div> |                                                                                                                                                                                                                                               |
| <div>DIFFERENTIATION ANGLE</div> <div>Not computed</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                               |

## Not computed

|         |                                                                        |
|---------|------------------------------------------------------------------------|
| Product | aroTHERM plus Commercial / aroTHERM Split (scaled commercial variants) |
|---------|------------------------------------------------------------------------|

|         |                                                                                                                                                                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pricing | Published trade pricing for the aroTHERM plus residential range starts around £2,500–£3,500 at distribution for sub-16kW units; commercial-tier pricing for 30kW-class units is not publicly listed but industry sources suggest a significant uplift. |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |              |
|------------|--------------|
| Tech Specs | Not computed |
|------------|--------------|

### STRENGTHS

Vaillant has demonstrated R290 manufacturing competence at scale, Strong UK brand and merchant distribution infrastructure, Recognised MCS certification on residential R290 products, Integration with existing Vaillant boiler replacement ecosystem

### WEAKNESSES

Commercial-specific R290 hydronic split offering in the 30kW bracket is not clearly defined in public product documentation as of knowledge cutoff, Primary positioning remains residential; commercial channel support may be thinner, Installer training for R290 commercial units may lag residential certification volumes, Indoor unit placement constraints for R290 may not be optimised for commercial environments in current product architecture

### DIFFERENTIATION ANGLE

Not computed

## Not computed

|         |                                                   |
|---------|---------------------------------------------------|
| Product | Altherma 3 R F (and commercial Altherma variants) |
|---------|---------------------------------------------------|

|         |                                                                                                                                                                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pricing | Industry sources and installer forums suggest Daikin Altherma commercial units in the 20–30kW range are priced to installers in the region of £6,000–£10,000 depending on configuration, though verified list pricing is not publicly available. |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |              |
|------------|--------------|
| Tech Specs | Not computed |
|------------|--------------|

### STRENGTHS

Dominant UK market share in heat pump sector broadly, Comprehensive MCS and non-domestic certification portfolio, Extensive technical support, training centres, and spare parts availability, Strong specification preference among M&E consultants

### WEAKNESSES

R32 refrigerant strategic vulnerability under accelerating F-Gas regulation, Premium pricing structure may be misaligned with small commercial budget constraints, Product complexity and BMS integration requirements may deter smaller installers, Less agile than specialist manufacturers in responding to niche segment requirements

### DIFFERENTIATION ANGLE

Not computed

## Not computed

Product

GMLW / Europa series air-source heat pumps

Pricing

Ochsner commercial heat pump pricing is not widely published in UK distribution channels; European market data suggests commercial units in this power class are positioned at a premium, likely £8,000–£15,000 equivalent at installed equipment cost.

Tech Specs

Not computed

### STRENGTHS

Strong engineering heritage in commercial heat pump design, High-quality build specification with long operational life expectation, Experience with demanding commercial and light industrial applications

### WEAKNESSES

Limited UK distribution and installer ecosystem, MCS certification status for UK non-domestic market requires verification, Premium pricing limits addressable market in cost-sensitive small commercial segment, Brand recognition among UK M&E specifiers is lower than major Japanese or German competitors

### DIFFERENTIATION ANGLE

Not computed

## Not computed

Product

NIBE F2040 / F2120 air-source heat pump range

Pricing

Published UK trade sources and installer quotes suggest NIBE commercial-class air-source units in the 20–30kW bracket are priced in the £5,000–£9,000 range at distribution, with variation by configuration.

Tech Specs

Not computed

### STRENGTHS

Established UK non-domestic MCS certification on relevant models, Good reputation in commercial and light industrial hydronic applications, Integration with NIBE SMO control platform and BMS connectivity, Active UK distributor network with technical support

### WEAKNESSES

R410A models face near-term obsolescence risk; R32 transition ongoing, No R290 offering in this power class identified in current public documentation, Pricing at the upper end of small commercial budget tolerance, Product range breadth may result in less focused small commercial sales support

### DIFFERENTIATION ANGLE

Not computed

### Data Sources

[LLM Search] Aggregated competitor data via LLM

**Overall Section Grade: D**



# 1.4 Constraints & Targets <sup>[A]</sup>

|                     |              |
|---------------------|--------------|
| Target Material     | Not computed |
| Target Process      | Not computed |
| Tolerance Target    | Not computed |
| Quantity Target     | Not computed |
| Batch Size          | 500          |
| Target Cost Ceiling | £30.00       |
| Max Mass            | Not computed |

**Data Sources**

[Mixed] Constraints provided by User, Sources generated by LLM

**Overall Section Grade: B**

## 2. Regulatory & Compliance

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No regulatory standards identified for this brief.

### 3. Sizing & Spatial Optimisation <sup>[D]</sup>

#### INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

#### System Envelope

|                             |                       |
|-----------------------------|-----------------------|
| Internal Dimensions (W×D×H) | 5000 × 5000 × 3000 mm |
| Internal Floor Area         | 25 m <sup>2</sup>     |
| Internal Volume             | 75 m <sup>3</sup>     |
| Rules Domain                | hardware_product      |

#### Module Allocation Zone Table

| Module                                        | L × W × H (mm)     | Area (m <sup>2</sup> ) | Mount |
|-----------------------------------------------|--------------------|------------------------|-------|
| compressor_drive_assembly                     | 2236 × 2236 × 2400 | 5                      | floor |
| outdoor_evaporator_fan_system                 | 2236 × 2236 × 2400 | 5                      | floor |
| r290_refrigerant_management_safety            | 2236 × 2236 × 2400 | 5                      | floor |
| indoor_brazed_plate_heat_exchanger_condenser_ | 2236 × 2236 × 2400 | 5                      | floor |
| hydraulic_circulation_control_module          | 2236 × 2236 × 2400 | 5                      | floor |
| primary_microprocessor_controls               | 2236 × 2236 × 2400 | 5                      | floor |
| 400v_3_phase_power_distribution               | 2236 × 2236 × 2400 | 5                      | floor |
| outdoor_casing_acoustic_shielding             | 2236 × 2236 × 2400 | 5                      | floor |

### Conflicts

- Total module footprint (40.0m<sup>2</sup>) exceeds envelope budget (21.3m<sup>2</sup>)

### Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

**Overall Section Grade: B**

# 4.1 Compressor & Drive Assembly [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapor into high-pressure, high-temperature gas to drive the heating cycle. It modulates speed via an inverter to accurately match the building's 30kW thermal load.

Why It Matters

Not computed

Technical Description

This module features an advanced inverter-driven scroll or twin-rotary compressor specifically optimized for R290 refrigerant. It operates within a hermetically sealed environment entirely contained within the outdoor unit to comply with ATEX and EN 378 safety standards, minimizing flammability risks. By operating exclusively outdoors, the volatile A3 refrigerant is separated from the interior occupied spaces.

Because R290 possesses high latent heat and excellent low-temperature performance characteristics, the compressor can achieve a 30kW thermal capacity at A2/W35 while maintaining a COP of 3.5 or higher. It draws approximately 8.5kW to 9.0kW of maximum electrical input. The integrated inverter drive is crucial for limiting peak starting currents, allowing the system to smoothly modulate against the commercial heating load without violating grid constraints.

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.2 Outdoor Evaporator & Fan System [D]

Maturity: draft

Purpose

Extracts strictly ambient heat from the outside air by blowing it across a large fin-and-tube coil. It includes the necessary hardware for shedding ice during cold-weather defrost protocols.

Why It Matters

Not computed

Technical Description

The outdoor evaporator utilizes a massive fin-and-tube coil array, heavily surface-treated with an epoxy or hydrophilic coating to resist harsh UK weather and coastal corrosion. R290 evaporates rapidly inside these copper or aluminum tubes, absorbing freely available atmospheric heat and boiling at exceptionally low temperatures, maintaining capability even down to -20°C ambient conditions.

Two variable-speed EC (Electronically Commutated) fan motors, optimally specified as IE4 efficiency class, force massive volumes of air across the coil. Low-noise operation is prioritized to keep the unit below the 60 dB(A) acoustic threshold required for tight commercial-site planning boundaries. A built-in reverse-cycle or hot gas bypass system handles automatic defrosting to quickly clear transient ice accumulation in high-humidity winter environments.

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**



# 4.3 R290 Refrigerant Management & Safety [D]

Maturity: draft

Purpose

Meters the flow of the highly flammable A3 liquid propane to maximize thermal lift, while housing all mission-critical safety and leak detection components.

Why It Matters

Not computed

Technical Description

This sub-system controls the expansion and flow of the 1.5-4 kg R290 charge using an Electronic Expansion Valve (EEV). The EEV rapidly modulates to maintain perfect target superheat, maximizing heat extraction from the evaporator while preventing raw liquid from returning to the compressor. All proximate electrical components within the enclosure are ATEX-rated or hermetically sealed to eliminate potential ignition sources.

Because R290 is an A3 highly flammable gas, this subsystem includes stringent safety mechanisms tailored to EN 378 and EN 60335-2-40 standards. It incorporates active leak detection sensors that instantly shut off the compressor, isolate the refrigerant flow, and continuously purge the system enclosure via the primary fans if a micro-leak is detected. This redundant safety layer ensures absolute safety for demanding commercial environments.

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.4 Indoor Brazed Plate Heat Exchanger (Condenser) [D]

Maturity: draft

## Purpose

Transfers intense thermal energy from the high-pressure R290 gas into the commercial building's water-glycol heating circuit. It serves as the bridge between the refrigerant cycle and the building's hydronics.

## Why It Matters

Not computed

## Technical Description

The Brazed Plate Heat Exchanger (BPHE), situated inside the indoor hydronic module, serves as the central condenser during standard heating mode. Hot, high-pressure R290 gas enters one side of the stacked stainless-steel plate series, while cooler system water enters the opposing parallel channels. The massive latent heat of the R290 transfers efficiently into the water, raising its temperature substantially without requiring highly inefficient secondary electrical backup.

The guaranteed physical separation of these circuits is vital. By keeping the complete refrigeration circuit physically isolated between the outdoor unit and this primary heat exchanger, the inherently hazardous R290 never enters the occupied commercial building. The BPHE exterior is meticulously insulated with EPDM foam to strictly prevent thermal loss and avoid condensation pooling when utilizing optional summer cooling modes.

## Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.5 Hydraulic Circulation & Control Module [D]

Maturity: draft

Purpose

Circulates the heated water/glycol mixture into the building's commercial distribution system. It maintains appropriate system pressure, accommodates thermal expansion, and buffers thermal output.

Why It Matters

Not computed

Technical Description

Located entirely within the indoor module, this system incorporates a heavy-duty variable-speed circulation pump (typically sourced from Grundfos or Wilo) engineered for substantial commercial flow rates (approximately 5.1 m³/hr at a 5K delta-T). It dictates the precise rate at which heat is pulled away from the heat exchanger and pushed toward the commercial emitters or domestic hot water cylinder feeds.

The subsystem is fully equipped for complete closed-loop hydronic management out of the box. It houses a composite or stainless-steel buffer vessel designed to prevent the compressor from short-cycling during low thermal load intervals. An integrated expansion vessel absorbs hydraulic pressure changes caused by significant temperature fluctuations, whilst a mechanical safety relief valve aggressively prevents dangerous over-pressurization of the pipework.

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.6 Primary Microprocessor & Controls [D]

Maturity: draft

Purpose

Acts as the strategic brain, calculating weather compensation, matching inverter speeds to commercial thermal loads, and communicating seamlessly with external facility networks.

Why It Matters

Not computed

Technical Description

This advanced microprocessor board sequences the entire heat pump methodology, from the cautious soft-start of the 400V inverter to the micro-adjustments of the EEV. It continuously monitors external ambient temperature, return water temperature, and compressor discharge pressure in real-time to precisely calculate the optimal weather compensation curve. This strict monitoring ensures the unit dynamically produces only the heat required by the building, exponentially maximizing seasonal efficiency thresholds.

Crucially for the UK commercial market (schools, offices, clinics), this control module features incredibly deep integration capabilities out of the factory. It incorporates a direct, native gateway for Modbus RTU, Modbus TCP, and BACnet protocols. This architecture allows facilities managers to granularly monitor performance, remotely clear faults, or easily integrate the heat pump into broader cascaded heating systems (efficiently managing multiple 30kW units acting together).

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**



# 4.7 400V 3-Phase Power Distribution [D]

Maturity: draft

## Purpose

Safely receives, filters, and distributes 3-phase commercial grid power to the heavy-draw components, carefully managing maximum electrical input and harmonic distortion.

## Why It Matters

Not computed

## Technical Description

This electrical module handles the substantial and demanding electrical requirements of a 30kW commercial heat pump, taking in a designated 400V, 50Hz 3-phase+Neutral supply. It continuously processes incoming power through heavy-duty EMI/RFI noise filters to comply stringently with European and UK electromagnetic compatibility directives. This filtering fundamentally prevents the inverter's high-frequency switching operations from polluting and destabilizing the commercial building's wider electrical grid.

Critically, instead of applying jarring Direct-On-Line (DOL) starts which create massive and destructive current spikes, this module carefully routs power through an advanced soft-start inverter bridge. This restricts the starting current to remain well below strict DNO limitations, facilitating far easier G99 network operator clearance for installers. The module crucially includes robust 3-phase AC-3 contactors and fast-acting circuit breakers to aggressively physical isolate the compressor and auxiliary heaters immediately during a severe fault.

## Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.8 Outdoor Casing & Acoustic Shielding [D]

Maturity: draft

Purpose

Protects all internal components from severe weather and salt spray while absorbing brutal compressor and fan vibrations to strictly limit acoustic emissions.

Why It Matters

Not computed

Technical Description

The major outdoor unit's structural chassis is constructed intricately from heavy-gauge galvanized sheet steel, rigorously treated with a C4-class protective powder coating designed specifically to withstand the harsh, damp UK climate and penetrating coastal salinity. The internal framing is highly engineered for maximum structural rigidity, providing a stable, unyielding foundation for the 200kg+ mass of the heat exchanger and operating compressor suite, optimally accommodating typical dual-fan vertical or large twin-fan horizontal discharge profiles.

Acoustics represent a significant historical barrier to achieving planning permission in the UK construction sector. This module aggressively combats noise pollution through a multi-layered acoustic compressor jacket, dense acoustic foam permanently adhering to the interior of external panels, and heavy-duty elastomeric anti-vibration mounts securely isolating the compressor from the metal floor plate. These combined features comprehensively strip away low-frequency mechanical rumbling and high-frequency aerodynamic fan whine, successfully bringing sound pressure profiles well down to permissible commercial levels.

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 5. Cost Waterfall & Economics [D]

## Unit Cost Breakdown

| Module              | Subtotal     |
|---------------------|--------------|
| Total Unit BOM Cost | Not computed |

## Non-Recurring Engineering (NRE)

| Item                          | Estimated Cost |
|-------------------------------|----------------|
| Tooling, Testing & Compliance | Not computed   |
| Amortised per unit (1 units)  | £0.00          |

## Cost Reduction Paths

| Strategy                              | Est. Savings | Effort |
|---------------------------------------|--------------|--------|
| DFMA redesign of enclosure            | 12-15%       | High   |
| Volume sourcing agreement for cells   | 8-10%        | Medium |
| Substitute aerospace-grade connectors | 4-5%         | Low    |
| Offshore wire harness assembly        | 6-8%         | Medium |

**Data Sources**

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

**Overall Section Grade: C**

## 6. Failure Modes & Effects Analysis (FMEA) <sup>[D]</sup>

| Module                               | Hazard       | Cause        | Effect       | S×L=RPN | Mitigation & Verification Test                                                                                 |
|--------------------------------------|--------------|--------------|--------------|---------|----------------------------------------------------------------------------------------------------------------|
| Compressor & Drive Assembly          | Not computed | Not computed | Not computed | 5×2=10  | Implement strictly governed superheat control logic and high-capacity suction accumulators.                    |
| Compressor & Drive Assembly          | Not computed | Not computed | Not computed | 4×2=8   | Utilize oversized heat sinks, thermal pastes, and physical cut-outs on the inverter drive casing.              |
| Compressor & Drive Assembly          | Not computed | Not computed | Not computed | 3×3=9   | Enforce programmed oil-return cycles at maximum RPM every few operational hours.                               |
| Outdoor Evaporator & Fan System      | Not computed | Not computed | Not computed | 4×2=8   | Deploy dual-redundant temperature sensors for accurate dynamic defrost initiation.                             |
| Outdoor Evaporator & Fan System      | Not computed | Not computed | Not computed | 3×2=6   | Specify IE4-rated sealed-bearing EC fans with integrated thermal overload protection.                          |
| Outdoor Evaporator & Fan System      | Not computed | Not computed | Not computed | 3×4=12  | Standardize C4-rated corrosion protection and epoxy-treated coils for all UK SKUs.                             |
| R290 Refrigerant Management & Safety | Not computed | Not computed | Not computed | 5×2=10  | Design for minimal brazed joints and strictly mandate helium mass-spectrometer leak testing during production. |
|                                      |              |              |              |         |                                                                                                                |

|                                                       |              |              |              |               |                                                                                                                          |
|-------------------------------------------------------|--------------|--------------|--------------|---------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>R290 Refrigerant Management &amp; Safety</b>       | Not computed | Not computed | Not computed | <b>3×3=9</b>  | Install a high-capacity filter drier upstream of the EEV and use highly durable stepper motors.                          |
| <b>R290 Refrigerant Management &amp; Safety</b>       | Not computed | Not computed | Not computed | <b>2×3=6</b>  | Calibrate sensors uniquely to the strict spectral signature of R290 and shield them from ambient pollutants.             |
| <b>Indoor Brazed Plate Heat Exchanger (Condenser)</b> | Not computed | Not computed | Not computed | <b>5×1=5</b>  | Incorporate low-flow anti-freeze protection logic and strictly mandate installation of in-line magnetic filters.         |
| <b>Indoor Brazed Plate Heat Exchanger (Condenser)</b> | Not computed | Not computed | Not computed | <b>3×4=12</b> | Explicitly require installers to flush systems chemically and install heavy-duty dirt separators prior to commissioning. |
| <b>Indoor Brazed Plate Heat Exchanger (Condenser)</b> | Not computed | Not computed | Not computed | <b>4×2=8</b>  | Use highly reliable solid-state paddle flow switches combined with redundant pressure differential sensors.              |
| <b>Hydraulic Circulation &amp; Control Module</b>     | Not computed | Not computed | Not computed | <b>3×3=9</b>  | Include reliable automated air-venting valves and highly structured commissioning walk-throughs in the manual.           |
| <b>Hydraulic Circulation &amp; Control Module</b>     | Not computed | Not computed | Not computed | <b>2×3=6</b>  | Generously oversize the primary expansion vessel and utilize only high-grade EPDM rubber bladders.                       |
| <b>Hydraulic Circulation &amp; Control Module</b>     | Not computed | Not computed | Not computed | <b>4×2=8</b>  | Mechanically wire immersion backup through physical high-limit thermostats and demand physical flow verification.        |
| <b>Primary Microprocessor &amp; Controls</b>          | Not computed | Not computed | Not computed | <b>4×1=4</b>  | Implement dual-bank memory for firmware with strict auto-rollback protocols on watchdog timeout.                         |

|                                                |              |              |              |              |                                                                                                                      |
|------------------------------------------------|--------------|--------------|--------------|--------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Primary Microprocessor &amp; Controls</b>   | Not computed | Not computed | Not computed | <b>2×4=8</b> | Provide a highly intuitive installer touch-screen interface with built-in network node diagnostics.                  |
| <b>Primary Microprocessor &amp; Controls</b>   | Not computed | Not computed | Not computed | <b>3×3=9</b> | Utilize high-precision PT1000 thermistors combined with routine software offset self-calibration routines.           |
| <b>400V 3-Phase Power Distribution</b>         | Not computed | Not computed | Not computed | <b>4×2=8</b> | Incorporate intelligent automatic phase-loss detection with a timed auto-restart upon stable grid voltage return.    |
| <b>400V 3-Phase Power Distribution</b>         | Not computed | Not computed | Not computed | <b>4×2=8</b> | Specify premium-grade commercial contactors natively rated for harsh AC-3 inductive electrical loads.                |
| <b>400V 3-Phase Power Distribution</b>         | Not computed | Not computed | Not computed | <b>3×1=3</b> | Use high-temperature rated, long-life industrial DC link capacitors in the central inverter design.                  |
| <b>Outdoor Casing &amp; Acoustic Shielding</b> | Not computed | Not computed | Not computed | <b>3×3=9</b> | Consistently apply architectural standard C4 powder-coating with optional C5 marine factory upgrades.                |
| <b>Outdoor Casing &amp; Acoustic Shielding</b> | Not computed | Not computed | Not computed | <b>2×2=4</b> | Utilize heavy closed-cell acoustic insulation highly resistant to moisture absorption and UV structural degradation. |
| <b>Outdoor Casing &amp; Acoustic Shielding</b> | Not computed | Not computed | Not computed | <b>2×4=8</b> | Implement solid panel stiffeners and robust rubber edge gaskets aggressively on all removable service doors.         |



### Data Sources

[LLM Synthesis] FMEA Matrix Generation

**Overall Section Grade: E**

# 7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

| Section                   | Grade | Primary Source Type            |
|---------------------------|-------|--------------------------------|
| 1. Brief & Requirements   | B     | User Input & Verified Search   |
| 2. Regulatory Posture     | D     | LLM Knowledge Base             |
| 3. Sizing Optimization    | B     | Deterministic Solver           |
| 4. System Modules & Specs | D     | Algorithmic Extraction + LLM   |
| 5. Economics Waterfall    | C     | Arithmetic Model + Estimations |
| 6. FMEA Risk              | E     | Pure LLM Synthesis             |

# 8. Audit Log

Pipeline execution trace for traceability and verification.

| #  | Phase        | Action                             | Result                          |
|----|--------------|------------------------------------|---------------------------------|
| 1  | Ingest       | Parsed unstructured text brief     | Extracted variables             |
| 2  | Research     | Scanned for market and competition | Built landscape model           |
| 3  | Regulatory   | Mapped regional standards          | Identified 5-10 standard codes  |
| 4  | Architecture | Decomposed to functional modules   | 8 modules created               |
| 5  | Sizing       | Algorithmic 3D packing             | Constraints failed              |
| 6  | BOM          | Synthesised constituent parts      | 0 lines generated               |
| 7  | Sourcing     | Matched parts to supplier APIs     | Selected preferred vendors      |
| 8  | Economics    | Applied cost models                | Computed Not computed           |
| 9  | Review       | Cross-specialist critique          | Addressed engineering conflicts |
| 10 | Risk         | Populated FMEA table               | Computed RPN values             |
| 11 | Publish      | Generated final PDF report         | Success                         |